DATE CLEANING BASICS

Task 27 | Data Analytics Internship

Veda Technology

1. Objective

The objective of this task is to standardize inconsistent date formats in a retail sales dataset, identify invalid and missing dates, and keep a record of entries that need further review.

2. Tools and Dataset

Tools: Python, Pandas, Microsoft Excel
Dataset: Synthetic retail sales dataset created for practice
Records processed: 25

3. Cleaning Method

The Python script parses mixed date strings using Pandas. It uses a day-first convention for ambiguous numeric dates and converts values that cannot be parsed into a missing datetime value. Blank values are labelled Missing, while non-blank values that fail parsing are labelled Invalid.

4. Results

The script processed 25 sample records. Twenty-two dates were parsed successfully, two were identified as invalid, and one was missing. Invalid and missing values were flagged for review rather than being corrected by guesswork.

Status	Count	Meaning
Valid	22	Date was successfully parsed.
Invalid	2	Non-blank value could not be parsed as a valid date.
Missing	1	Date field was blank.

5. Issue Log

The issue log lists each invalid or missing date, its order ID, its status, and a recommended action. Invalid dates should be checked against the source record; missing dates should be recovered from the original source if possible.

6. Why Date Cleaning Matters

Inconsistent date formats can cause errors in sorting, filtering, date comparisons, and monthly or yearly reporting. Parsing dates into a consistent datetime type makes these operations more reliable. Ambiguous numeric dates must be interpreted using the source system's documented convention.

7. Interview Questions

Why are dates often problematic?
Dates may use different formats, regional conventions, invalid calendar values, or missing entries. These issues can lead to incorrect sorting and time-based analysis.

What is a date datatype?
A date or datetime datatype stores a date as a structured value rather than plain text, allowing date comparisons, sorting, calculations, and grouping.

8. Conclusion

This task provided practice in parsing inconsistent dates, separating invalid values from missing ones, and documenting data-quality issues. The cleaned dataset and issue log support further review before the data is used for analysis.

Prepared for Veda Technology | Task 27 – Date Cleaning Basics